



Course:	Object Oriented Programming (OOP)	Semester:	Spring 2026
Section:	BCS-2G	Total Marks:	100
Assignment - 1		Deadline:	15th Feb, 2026

Submission Instructions

(Please read the following instructions carefully before submitting the assignment).

1. Save your file with your Roll Number (format: xxL-xxxx_A1.cpp).
2. It should be clear that your assignment will not get any credit if:
 - The assignment is submitted after the due date.
 - Assignment is plagiarized from any source. (online/other students/etc.)

Important Notes:

- I. There should not be any memory leakage in your program, otherwise marks will be deducted.
- II. Your program must be as modular as possible. Your main() should be a sequence of function calls.
- III. You are not allowed to use global variables and go-to instruction.
- IV. All allocations of pointers should be dynamic. Use of static arrays is prohibited.
- V. Remember to delete the allocated memory when it is no longer needed.
- VI. All the data will be given by the user. Do not submit a program that runs on hard coded inputs.
- VII. Pass the pointers to functions instead of subscript notation [].
- VIII. Make proper functions to solve the problems.
- IX. Debug your code extensively to find errors/bugs. Test modules against the sample input given in the assignment.
- X. Use of "string" datatype and all built-in functions like strcmp() and strtok() etc. is strictly prohibited.
- XI. Write readable, well-formatted and well-commented code.
- XII. No matter how hard things seem, be confident and believe in your own abilities.

XIII. Do not panic and remember to take breaks from your device screen while attempting this assignment.

Task 1:

Write a program in C++ to implement dynamic memory allocation for a 2D maze game. The game should allow a player to move from one cell to another in the maze and reach the end. The player should be able to move in any of the four directions (up, down, left, right) as long as there is no wall in that direction. The maze should be randomly generated at the start of each game, using dynamic memory allocation. The game should keep track of the player's position and the number of moves taken to reach the end of the maze. The game should also check if the player has reached the end of the maze and display a path to the player indicating whether they won or lost the game. You need to start from top left and reach till bottom right.

Example 1:

```
1 1 0 0 0
0 1 1 1 0
1 1 0 1 0
0 0 1 1 0
0 0 1 1 1
```

Path 1 => right, down, right, right, down, down, left, down, right, right

Path 2 => right, down, right, right, down, down, down, right

Example 2:

```
1 1 0 0 0
0 1 1 1 0
1 1 0 1 0
0 0 1 0 0
0 0 1 1 1
```

Path => no path exists

Your task is to print shortest path as in first example is path 2.

Task 2: Dynamic Memory Based Student Quiz Score Tracker

You are required to design a quiz score tracking system using dynamic memory allocation in C++.

Problem Description:

Each student can attempt a quiz multiple time. For each student:

- Store the student's name
- Store the number of attempts
- Store the score of each attempt using dynamic memory allocation

Requirements:

Ask the user for the number of students. For each student:

- Dynamically allocate memory to store the student's name.

- Ask for number of quiz attempts.
- Dynamically allocate an array to store scores for each attempt.

Calculate and display:

Average score of each student

Best score of each student

Allow the user to:

Add one more attempt for any student (resize the score array dynamically).

Free all allocated memory properly at the end.

Constraints

No STL containers (vector, string)

Use new, delete

No global variables

Index bounds must be checked

Example Input / Output:

Enter number of students: 2

Student 1 name: Ali

Number of attempts: 3

Enter scores: 70 80 90

Student 2 name: Sara

Number of attempts: 2

Enter scores: 85 75

--- Results ---

Ali -> Average: 80 | Best: 90

Sara -> Average: 80 | Best: 85

Add another attempt for a student? (1 = Yes, 0 = No): 1

Enter student index (0-based): 1

Enter new score: 95

Updated Results:

Sara -> Average: 85 | Best: 95

BONUS: MAKE A VISUALLY APPEALING DISPLAY AND MENU